

Questions

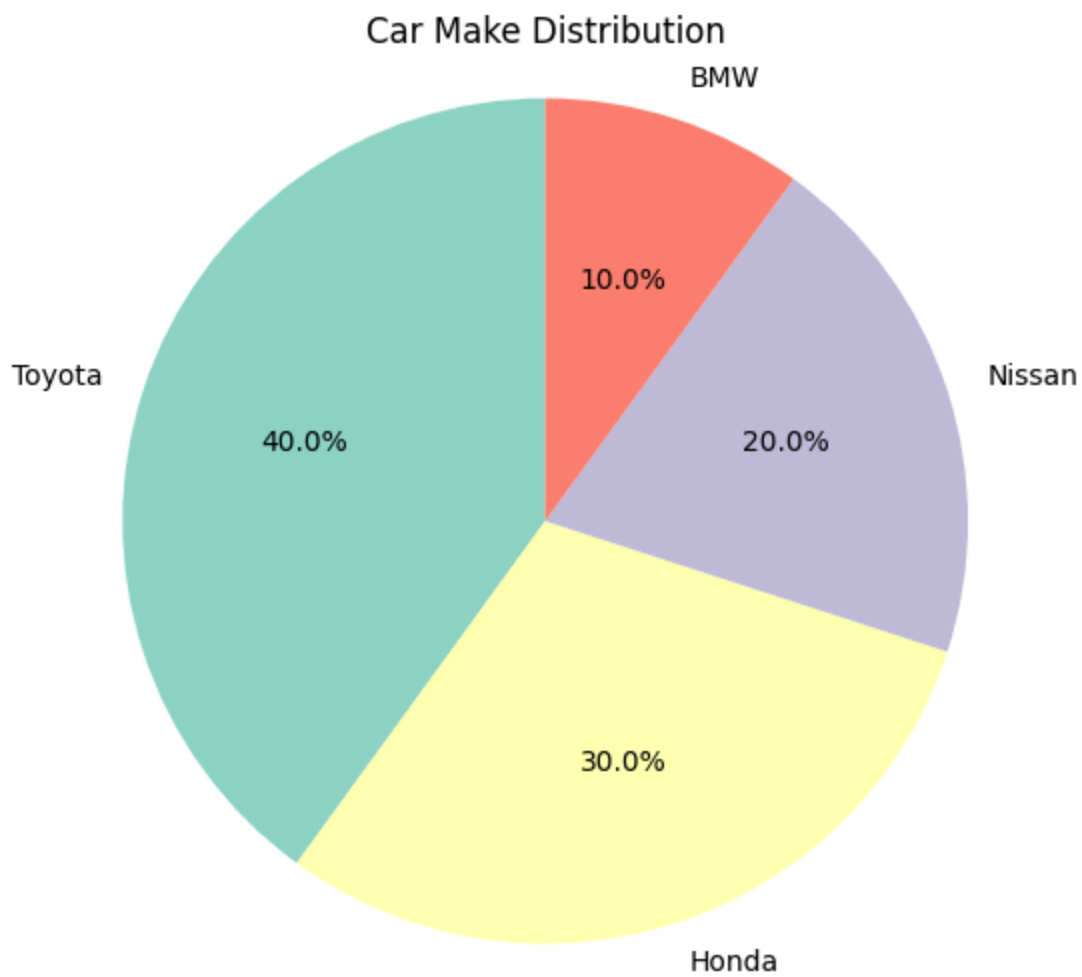
- 1- d (column chart)
 - 2- c (pie chart)
 - 3- d (box & whisker plot)
 - 4- b (histogram)
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Car Listings Dataset

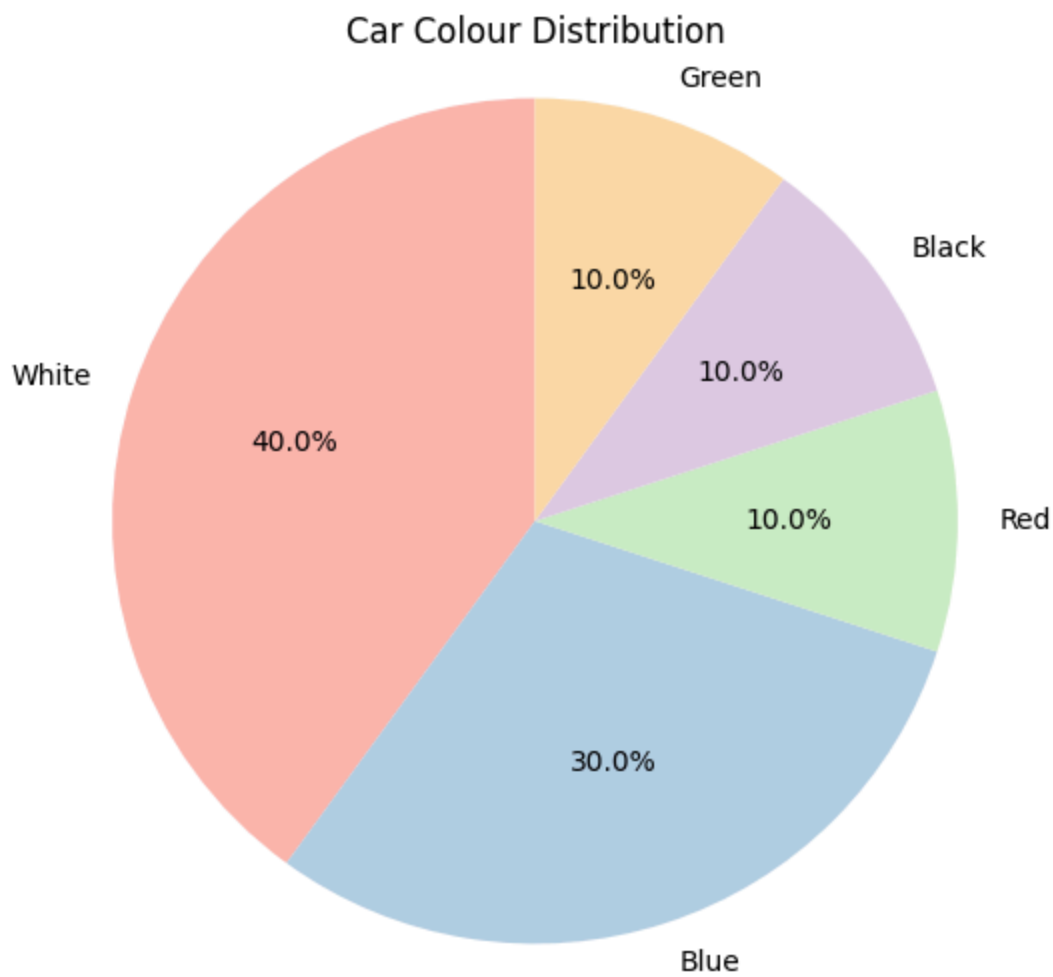
This dataset includes information about 10 used cars, providing details such as their make, color, odometer reading, number of doors, and price.

Dataset

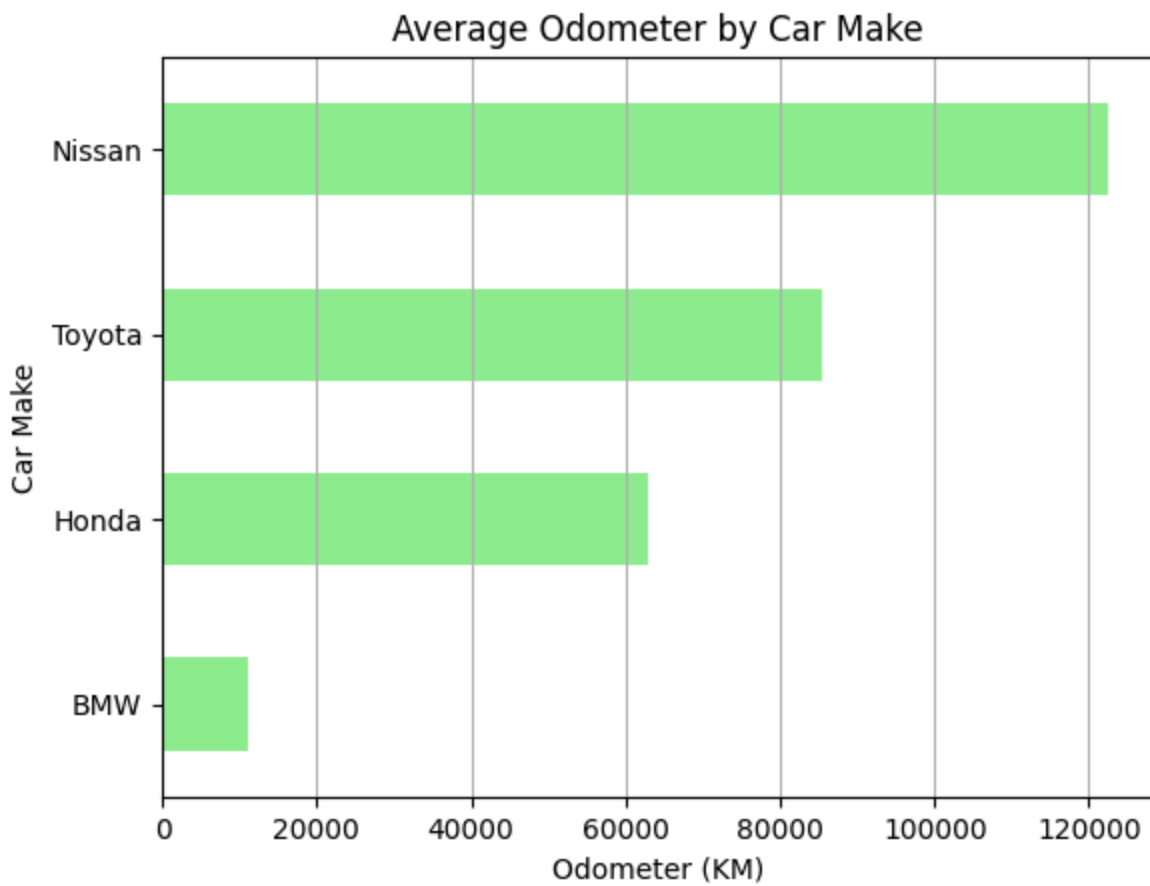
Index	Make	Colour	Odometer (KM)	Doors	Price
0	Toyota	White	150,043	4	\$4,000.00
1	Honda	Red	87,899	4	\$5,000.00
2	Toyota	Blue	32,549	3	\$7,000.00
3	BMW	Black	11,179	5	\$22,000.00
4	Nissan	White	213,095	4	\$3,500.00
5	Toyota	Green	99,213	4	\$4,500.00
6	Honda	Blue	45,698	4	\$7,500.00
7	Honda	Blue	54,738	4	\$7,000.00
8	Toyota	White	60,000	4	\$6,250.00
9	Nissan	White	31,600	4	\$9,700.00



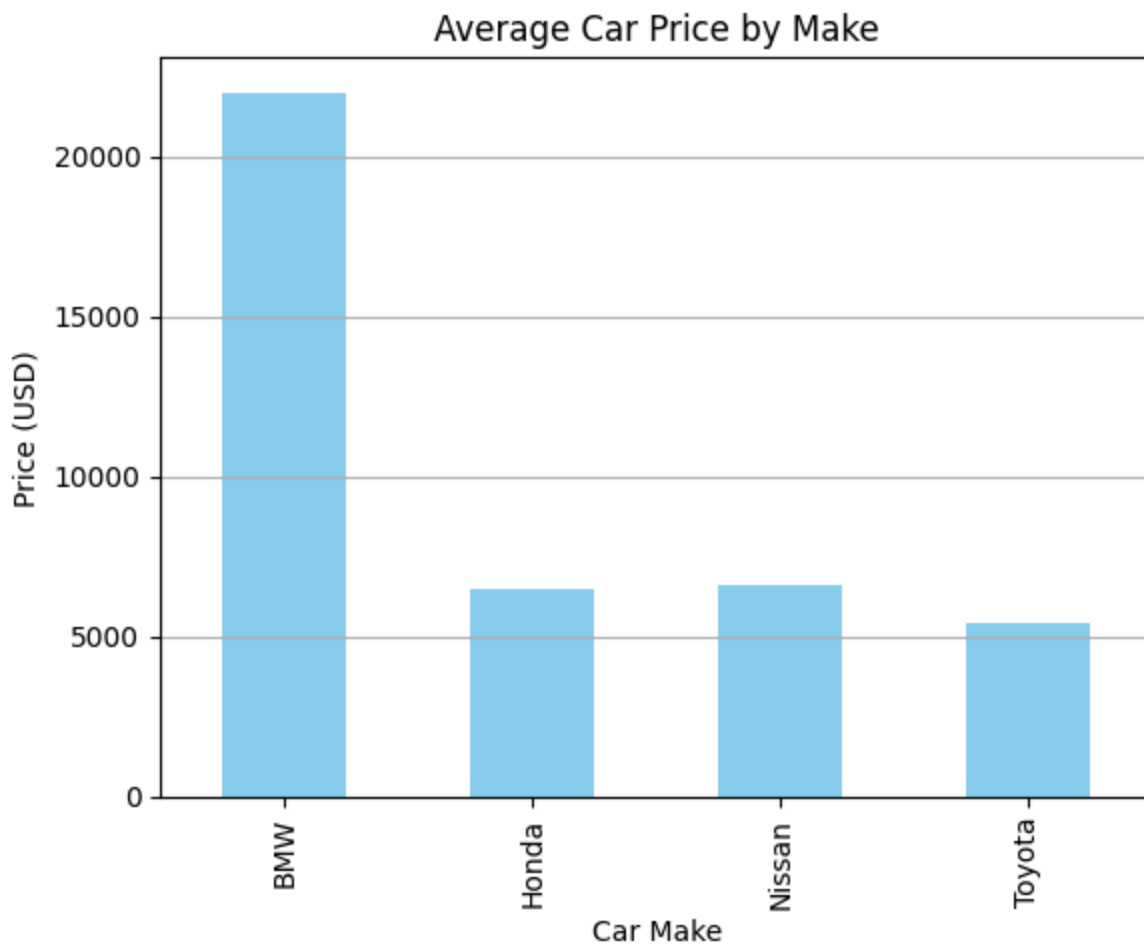
Question 1: What are the most common car colours?



Question 2: What is the relationship between car make and price?



Question 3: Which car make has the highest average odometer reading?



Which Data sets ?

which data sets have the same mean.

* Mean = $\frac{\text{sum of all values}}{\text{num. of values}}$

A) 9, 11, 15, 21, 7 Mean = $\frac{63}{5} = 12.6$

B) 12, 11, 14, 15, 9 Mean = $\frac{61}{5} = 12.2$

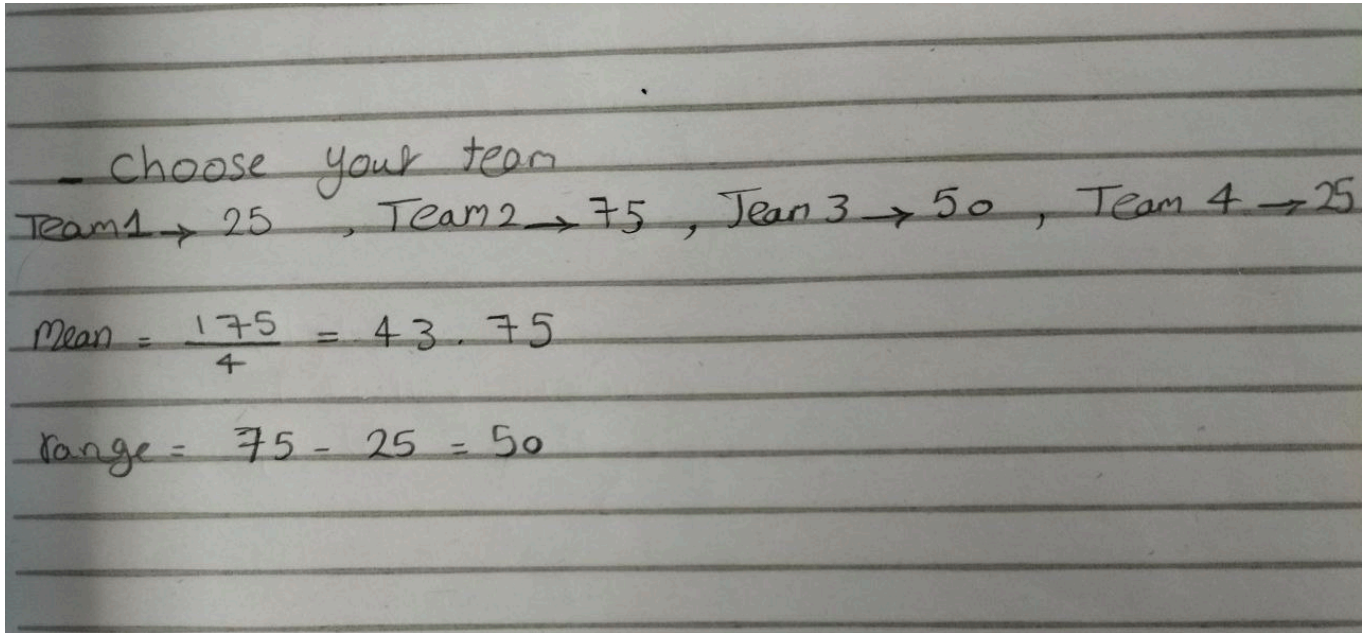
C) 7, 11, 30, 8, 9 Mean = $\frac{65}{5} = 13$

D) 6, 14, 10, 13, 22 Mean = $\frac{65}{5} = 13$

- answer C, D

- answer c , d

Choose your team



- answer
- mean = 43.75
- range = 50

Frequency Distribution Questions

1- Classes: **25–35, 35–45, 45–55, 55–65**

→ Frequencies: $2 + 5 + 3 + 2 = 12$

answer → b (12)

2- Classes: **35–45, 45–55, 55–65, 65–75**

→ Frequencies: $5 + 3 + 2 + 2 = 12$ workers

→ Percentage = $(12 / 20) \times 100 = 60\%$

answer → a (60%)

3- | **Arithmetic Mean** = $\frac{\sum xf}{\sum f}$:

Multiply midpoint × frequency and find the total:

Class	Midpoint	Freq	Mid × Freq
15–25	20	6	120
25–35	30	2	60
35–45	40	5	200
45–55	50	3	150
55–65	60	2	120
65–75	70	2	140

→ Total = $120 + 60 + 200 + 150 + 120 + 140 = \mathbf{790}$

→ Mean = $790 \div 20 = \mathbf{39.5}$

Answer → a (39.5)

4- There **could be** a spot in the lake where it is **deeper** than 1.5 meters.

Answer → (d)

Monthly Income Sample

Question1

1. **Range** = Max - Min = $40 - 3 = \mathbf{37}$

2. **Variance**

- Mean = $(40 + 3 + 3 + 4 + 4 + 6) / 6 = \mathbf{60 / 6 = 10}$

- Squared deviations from mean:

- $(40 - 10)^2 = 900$

- $(3 - 10)^2 = 49$

- $(3 - 10)^2 = 49$

- $(4 - 10)^2 = 36$

- $(4 - 10)^2 = 36$

- $(6 - 10)^2 = 16$

→ Sum = $900 + 49 + 49 + 36 + 36 + 16 = \mathbf{1086}$

- Variance = $1086 / 6 = \mathbf{181}$

3) **Standard deviation** = $\sqrt{181} \approx \mathbf{13.45}$

Question 2

If everyone scored **90**, there is no variation. (كده معنديش اي تنوع)

→ **Standard deviation = 0**

Answer → (d)

Question 3

If all values are negative, **measures of central tendency (like mean)** may be negative, but **dispersion (like standard deviation)** is always **non-negative**

Answer → (c)

Question 4

If **range = 0**, it means **all values are the same** → no variation.

If **median = 1**, then **all values = 1**

→ Mean = 1, SD = 0, mode = 1

Answer → (c)